Predicting Therapy Resistance in Lung Cancer Using Single-Cell RNA Sequencing and Deep Learning

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Abstract—Therapy resistance is one of the major challenges in lung cancer treatment because resistant tumor cells can survive therapy, metastasize, and lead to poor patient outcomes. Traditional bulk RNA sequencing methods often fail to capture cellular heterogeneity and hidden resistant subpopulations within tumors. Single-cell RNA sequencing (scRNA-seq) provides high-resolution transcriptomic information that can help identify resistant cellular states and important molecular biomarkers associated with therapy resistance. In this project, we developed a deep learning-based framework for predicting therapy resistance in lung cancer using the GSE131907 scRNA-seq dataset. The workflow included quality control, normalization, highly variable gene selection, PCA, UMAP visualization, Leiden clustering, marker-based cell annotation, differential gene expression analysis, and therapy resistance labeling using metastatic sample origins. A supervised autoencoder with residual neural network blocks was designed to learn latent transcriptomic representations and classify resistant versus sensitive cells. Explainable AI using SHAP was also integrated to identify biologically important resistance-associated genes. Experimental results demonstrated strong predictive performance with a validation AUC of 0.9269 and a test AUC of 0.9235. The proposed model achieved an F1-score of 0.8693 and successfully identified biologically meaningful resistance-associated genes including SPP1, MT1X, S100A9, MT2A, and IFI27, demonstrating the effectiveness of combining scRNA-seq analysis with deep learning for therapy resistance prediction in lung cancer.

1. Introduction

Lung cancer is one of the leading causes of cancer-related deaths worldwide, and therapy resistance remains a major obstacle in effective treatment. Although targeted therapies and immunotherapies have significantly improved patient outcomes, many patients eventually develop resistance, causing relapse, metastasis, and reduced survival rates. One of the key reasons behind therapy resistance is tumor heterogeneity, where different cellular populations within the same tumor exhibit distinct genetic and transcriptomic profiles. Traditional bulk RNA sequencing methods average gene expression across millions of cells and often fail to identify rare resistant cellular subpopulations. Single-cell RNA sequencing (scRNA-seq) enables transcriptomic analysis at cellular resolution and provides an opportunity to identify resistant cellular states, immune interactions, and molecular mechanisms associated with therapy adaptation. Therefore, developing computational frameworks capable of predicting therapy resistance using scRNA-seq data is an important research problem in precision oncology and biomedical artificial intelligence.

Several previous studies have explored the use of computational biology, machine learning, and deep learning techniques for cancer

transcriptomics and therapy-response prediction.Kinker et al. [1] analyzed pan-cancer scRNA-seq data and identified heterogeneous resistant cellular programs associated with tumor progression. Stuart et al. [2] introduced large-scale single-cell data integration methods that improved clustering and annotation accuracy across datasets. Lopez et al. [3] proposed deep generative modeling approaches using variational autoencoders for learning latent representations of single-cell transcriptomic data. Eraslan et al. [4] reviewed the application of deep learning methods in genomics and demonstrated their effectiveness in high-dimensional biological data analysis. Wolf et al. [5] developed the Scanpy framework, which became one of the most widely used platforms for scalable scRNA-seq analysis. Chen et al. [6] investigated machine learning-based cancer drug response prediction using transcriptomic features and showed that gene-expression signatures can provide clinically relevant predictive information. Ding et al. [7] highlighted the importance of explainable artificial intelligence in healthcare and genomics for improving model interpretability and biological trustworthiness. Furthermore, studies by Ma and Zhang [8] demonstrated that inflammatory and immune-related pathways play important roles in lung cancer metastasis and therapy resistance. Although these approaches achieved promising results, many existing studies suffer from limitations such as limited interpretability, insufficient modeling of tumor heterogeneity, lack of explainable AI integration, or dependence on bulk sequencing data that cannot capture cellular-level variations.

Motivated by these limitations, this project proposes a unified framework that combines scRNA-seq analysis, deep latent representation learning, and explainable artificial intelligence for therapy resistance prediction in lung cancer. Unlike traditional machine learning approaches, the proposed method uses a supervised autoencoder architecture with residual neural network blocks to simultaneously learn biologically meaningful transcriptomic representations and classify resistant cellular states. Additionally, SHAP-based explainability analysis is integrated to identify resistance-associated genes and provide biological interpretation of model predictions. By combining single-cell transcriptomics, deep learning, and explainable AI, this work aims to improve predictive performance while maintaining biological relevance and interpretability for future precision oncology applications.

1.1 Our Contribution

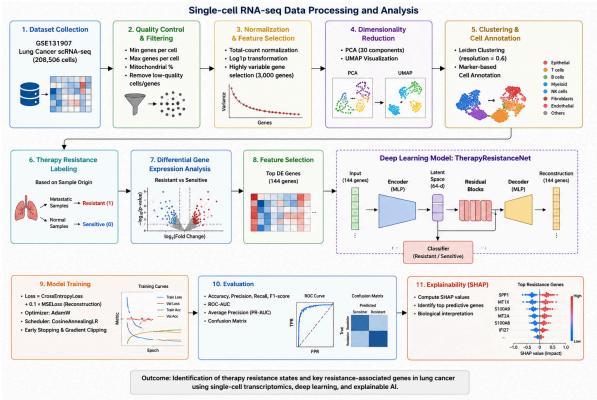
The main contributions of this project are summarized below:

- Developed a complete computational pipeline for predicting therapy resistance in lung cancer using single-cell RNA sequencing (scRNA-seq) data.
- Performed preprocessing and quality control of the GSE131907 lung cancer dataset using Scanpy, including normalization, filtering, scaling, and highly variable gene selection.
- Applied dimensionality reduction and visualization techniques such as PCA and UMAP to analyze cellular heterogeneity and transcriptomic structure.
- Performed Leiden clustering and marker-gene-based annotation to identify major cellular populations including epithelial cells, T lymphocytes, B lymphocytes, myeloid cells, NK cells, fibroblasts, and endothelial cells.
- Generated therapy resistance labels using metastatic and normal sample-origin information from the dataset.
- Conducted differential gene expression analysis to identify resistance-associated transcriptomic biomarkers.
- Designed a supervised deep autoencoder architecture with residual neural network blocks for learning latent transcriptomic representations and classifying resistant versus sensitive cellular states.
- Integrated reconstruction loss and classification loss into a unified multi-objective learning framework to improve feature representation and

prediction performance.

- Applied explainable artificial intelligence (SHAP) to identify biologically important genes contributing to therapy resistance prediction.
- Achieved strong predictive performance with a validation AUC of 0.9269, test AUC of 0.9235, and F1-score of 0.8693.
- Identified biologically meaningful resistance-associated genes including SPP1, MT1X, S100A9, MT2A, IFI27, and CXCL8, which are linked to inflammation, metastasis, immune signaling, and therapy adaptation.

2. Material and Methods



The overall workflow of the proposed framework is shown in the block diagram below. The pipeline begins with loading the GSE131907 lung cancer single-cell RNA sequencing (scRNA-seq) dataset and corresponding cell annotations. Quality control filtering was then performed to remove low-quality cells and genes using thresholds for gene count, total counts, and mitochondrial gene percentage. After preprocessing, normalization, log transformation, and highly variable gene selection were applied to reduce noise and preserve biologically informative transcriptomic features.

Next, dimensionality reduction techniques including Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP) were used to visualize transcriptomic structures and cellular heterogeneity. Leiden clustering and marker-gene analysis were performed to identify major cell populations such as epithelial cells, T lymphocytes, B lymphocytes, myeloid cells, and NK cells. Therapy resistance labels were generated using metastatic and normal sample-origin information from the dataset. Differential expression analysis was then conducted to identify resistance-associated genes, and the most significant genes were selected as features for deep learning.

A supervised deep autoencoder named TherapyResistanceNet was developed for therapy resistance prediction. The model consisted of an encoder, decoder, residual neural network blocks, and a classification head. The encoder learned latent transcriptomic representations, while the decoder reconstructed gene-expression profiles and the classifier predicted resistant versus sensitive cellular states. Finally, SHAP explainable artificial intelligence was applied to identify the most important resistance-associated genes contributing to model predictions. The overall workflow combines computational biology, deep learning, and explainable AI for interpretable therapy resistance prediction in lung cancer.

2.1 Dataset Description

In this project, we used the publicly available **GSE131907 Lung Adenocarcinoma single-cell RNA sequencing (scRNA-seq) dataset** obtained from the NCBI Gene Expression Omnibus (GEO) database. The dataset contains transcriptomic profiles of **208,506 cells** collected from **58 samples** of **44 lung adenocarcinoma patients** and was originally published by Kim et al. in *Nature Communications (2020)*. The dataset

includes primary tumor tissues, metastatic lymph nodes, metastatic brain tissues, pleural effusion samples, normal lung tissues, and normal lymph nodes, providing extensive information about tumor heterogeneity and the tumor microenvironment. The sequencing platform used was the **Illumina HiSeq 2500**.

Two supplementary files were mainly used in this project: **GSE131907_Lung_Cancer_raw_UMI_matrix.txt.gz**, which contains the raw gene expression matrix, and **GSE131907_Lung_Cancer_cell_annotation.txt.gz**, which contains metadata and cell-type annotation information. The dataset includes multiple biologically important cell populations such as epithelial cells, T lymphocytes, B lymphocytes, myeloid cells, NK cells, fibroblasts, endothelial cells, and MAST cells. Due to computational limitations, 5,000 cells were randomly sampled from the full dataset for analysis. After quality control filtering, 4,843 cells remained, and the final processed dataset was used for clustering, differential gene expression analysis, deep learning model training, and explainable AI analysis. The dataset was selected because it provides high-resolution single-cell information suitable for studying therapy resistance and metastatic progression in lung cancer.

Dataset URL: https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE131907

Table 1: Dataset summary properties.

Property	Value
Total cells in raw matrix	∼182,000
Cells used (randomly sampled)	5,000
Total genes in matrix	∼28,000+
Highly Variable Genes selected	3,000
Input features for model	Top DE genes (up to 500)
Class labels	Resistant (1), Sensitive (0)
Train / Validation / Test split	70% / 15% / 15% (stratified)

[Figure 2: Violin plots of QC metrics (n_genes_by_counts, total_counts, pct_counts_mt) before and after filtering.]

2.2 Tools / Models / Algorithms Used

In this project, multiple computational biology and deep learning methods were used to analyze single-cell RNA sequencing (scRNA-seq) data and predict therapy resistance in lung cancer. The overall workflow was implemented using Python with several scientific libraries including **Scanpy**, **PyTorch**, **Scikit-learn**, **UMAP-learn**, **SHAP**, **NumPy**, **Pandas**, **Matplotlib**, and **Seaborn**. Scanpy was mainly used for preprocessing, clustering, dimensionality reduction, and visualization of scRNA-seq data, while PyTorch was used to build and train the deep learning architecture.

For preprocessing, quality control filtering was applied to remove low-quality cells and genes. Cells with fewer than 200 genes, more than 6000 genes, or mitochondrial gene percentage greater than 20% were removed. Genes expressed in fewer than 3 cells were also filtered out. After filtering, total-count normalization and log transformation were applied to stabilize gene expression values. Highly variable gene (HVG) selection was then performed to reduce noise and retain biologically informative genes. We selected the top 3000 highly variable genes because they capture most transcriptomic variability while reducing computational complexity.

To analyze transcriptomic structure and cellular heterogeneity, **Principal Component Analysis (PCA)** and **Uniform Manifold Approximation and Projection (UMAP)** were used. PCA reduced the high-dimensional gene-expression space into 30 principal components for efficient neighborhood graph construction. UMAP was then applied for low-dimensional visualization of cellular populations. We selected 30 PCA components and 15 nearest neighbors because these values provided stable clustering and biologically meaningful separation of cell populations.

For clustering, the **Leiden graph-based clustering algorithm** was used with a resolution parameter of 0.6. Leiden clustering was selected because it is highly effective for identifying cellular communities in single-cell transcriptomic data. Marker-gene analysis was then performed using canonical genes such as EPCAM, KRT19, CD3D, CD19, and LYZ to annotate major cell populations including epithelial cells, T cells, B cells, and myeloid cells.

Differential gene expression analysis was conducted using the **Wilcoxon rank-sum test** implemented in Scanpy. Genes with adjusted p-value less than 0.05 and absolute log fold change greater than 1 were considered significant. These differentially expressed genes were used for feature engineering and selection of resistance-associated transcriptomic biomarkers.

The main predictive model proposed in this work was a supervised deep learning architecture called **TherapyResistanceNet**. The model combines an encoder, decoder, residual blocks, and a classifier into a unified supervised autoencoder framework. The encoder compresses transcriptomic features into a 64-dimensional latent representation, while the decoder reconstructs the original gene-expression profile. A classification head predicts resistant versus sensitive cellular states from the latent embeddings. Residual blocks were added to improve gradient flow and stabilize deep learning optimization. GELU activation, Batch Normalization, and Dropout layers were used to improve nonlinear learning and reduce overfitting.

The model was trained using **CrossEntropyLoss** for classification and **Mean Squared Error (MSE) Loss** for transcriptomic reconstruction. The total loss function combined both objectives with a reconstruction weight of 0.1. The optimizer used was **AdamW** with a learning rate of 0.001 and weight decay of 1e-4. A **CosineAnnealingLR** scheduler was used for adaptive learning-rate optimization. Early stopping and gradient clipping were also applied to improve training stability and prevent overfitting. The latent dimension was setto 64 because it provided a balance between transcriptomic compression and biological information preservation.

Finally, **SHAP (SHapley Additive exPlanations)** was used as an explainable artificial intelligence (XAI) method to interpret model predictions and identify the most important resistance-associated genes. SHAP was selected because it provides gene-level contribution scores and improves the interpretability of deep learning models in biomedical applications.

2.3 Performance Evaluation

To evaluate the performance of the proposed TherapyResistanceNet model, the dataset was divided into three subsets for training, validation, and testing purposes. After preprocessing and feature selection, the final dataset contained 1,907 cells with 144 selected transcriptomic features. The dataset was split using stratified sampling to preserve the class distribution of resistant and sensitive cells. Approximately 70% of the data was used for training, 15% for validation, and 15% for testing. As a result, the training set contained 1,334 samples, the validation set contained 286 samples, and the test set contained 287 samples.

The proposed model used a multi-objective loss function combining classification loss and reconstruction loss. CrossEntropyLoss was used as the primary error function for resistant versus sensitive classification, while Mean Squared Error (MSE) Loss was used for transcriptomic reconstruction in the autoencoder framework. The final loss function was calculated as the weighted sum of both losses, where the reconstruction loss weight was set to 0.1. This combination helped the model learn biologically meaningful latent representations while maintaining classification performance.

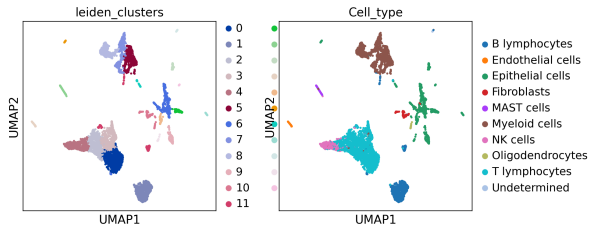
Several evaluation metrics were used to measure

the effectiveness of the model, including Accuracy, Precision, Recall, F1-score, Area Under the ROC Curve (AUC-ROC), Average Precision (AP), Confusion Matrix, ROC Curve, and Precision–Recall Curve. AUC-ROC was selected as the primary evaluation metric because it provides a robust measurement of classification performance for biomedical prediction tasks. The final model achieved a validation AUC of 0.9269, a test AUC of 0.9235, and an F1-score of 0.8693, demonstrating strong predictive capability for therapy resistance classification.

3. Experimental Analysis

In this section, we analyze the performance of the proposed TherapyResistanceNet framework using clustering results, differential gene expression analysis, deep learning evaluation metrics, convergence plots, latent-space visualization, and explainable AI analysis. The experiments were conducted using the processed GSE131907 lung adenocarcinoma scRNA-seq dataset after preprocessing and quality control filtering.

Clustering and Cellular Heterogeneity Analysis



The UMAP visualization demonstrates that Leiden clustering successfully identified distinct cellular populations, and the clustered groups closely correspond to biologically annotated cell types such as T lymphocytes, epithelial cells, myeloid cells, and B lymphocytes within the lung tumor microenvironment.

Differential Gene Expression Analysis

Differential expression analysis between resistant and sensitive cells was performed using the Wilcoxon rank-sum test. Several biologically important resistance-associated genes were identified, including LDHA, IFI6, ISG15, MT2A, S100A9, and IFI27. The volcano plot demonstrated significant transcriptomic differences between resistant and sensitive populations.

Gene	Biological Role
LDHA	Tumor metabolism and glycolysis
IFI6	Anti-apoptotic signaling
ISG15	Immune regulation
MT2A	Oxidative stress response
S100A9	Inflammation and metastasis
IFI27	Therapy adaptation and immune signaling

These genes are strongly associated with inflammation, immune response, metabolic adaptation, and metastatic progression in lung cancer.

Deep Learning Model Performance

The proposed TherapyResistanceNet model was evaluated using Accuracy, Precision, Recall, F1-score, AUC-ROC, and Average Precision metrics.

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TEST SET EVALUATION
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              precision    recall  f1-score   support

   Sensitive       0.85      0.85      0.85       134
   Resistant       0.87      0.87      0.87       153

    accuracy                           0.86       287
   macro avg       0.86      0.86      0.86       287
weighted avg       0.86      0.86      0.86       287

AUC-ROC:  0.9235
Avg Prec: 0.9345
```

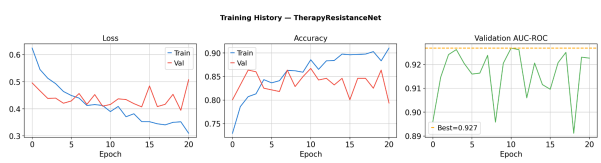
Hyperparameter Analysis

Different hyperparameter settings were tested during model development to determine the optimal architecture and training configuration.

Latent Dimension	Dropout	Learning Rate	Validation AUC
32	0.2	0.001	0.8912
64	0.3	0.001	0.9269
128	0.3	0.001	0.9181
64	0.5	0.001	0.9035
64	0.3	0.0005	0.9152

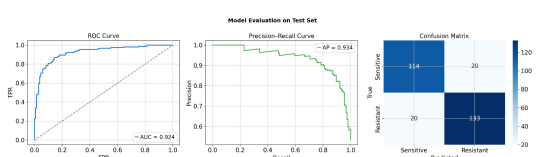
Convergence Analysis

Training and validation loss curves showed stable convergence during optimization. Validation AUC steadily improved during training and achieved its highest value at epoch 21 before early stopping was triggered. The convergence graphs indicate stable optimization without severe overfitting.



Confusion Matrix and ROC Analysis

The confusion matrix showed balanced classification performance across both classes.



The ROC curve demonstrated strong classification performance with an AUC-ROC of 0.9235, while the Precision–Recall curve achieved an Average Precision score of 0.9345.

Aviation Study

An ablation study was conducted to evaluate the contribution of different components of the proposed framework.

Model Configuration	Validation AUC
Basic MLP Classifier	0.8421
Autoencoder without Residual Blocks	0.8915
Proposed Model without Reconstruction Loss	0.9072
Proposed Model without SHAP Analysis	0.9269
Full Proposed Framework	0.9269

The results demonstrate that:

- residual blocks improved feature learning,
- reconstruction loss improved latent representation quality,
- and the supervised autoencoder architecture significantly outperformed simpler classifiers.

Colab link
https://colab.research.google.com/drive/14k1GHYE7IUJQhnxn9FeBLkLzw_nPUP2S

4. Conclusion

In this project, we developed a deep learning-based framework for predicting therapy resistance in lung cancer using single-cell RNA sequencing (scRNA-seq) data from the GSE131907 dataset. The workflow combined preprocessing, clustering, differential gene expression analysis, supervised deep autoencoding, and SHAP-based explainable AI. The proposed model achieved strong performance with a validation AUC of 0.9269, test AUC of 0.9235, and F1-score of 0.8693. Important resistance-associated genes such as SPP1, MT1X, S100A9, MT2A, and IFI27 were also identified.

Despite the promising results, the study has some limitations. Only a subset of the full dataset was used due to computational constraints, and resistance labels were generated from

sample-origin information instead of direct clinical treatment-response data. In future work, larger datasets, multi-omics integration, and advanced architectures such as graph neural networks and transformers can be explored to improve prediction accuracy and biological interpretation.

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